Hệ tọa độ Đề-các:

Gradient: $grad(f) = \nabla f = \frac{\partial f}{\partial x}\mathbf{a}_x + \frac{\partial f}{\partial y}\mathbf{a}_y + \frac{\partial f}{\partial z}\mathbf{a}_z$ Độ tản: $div(\mathbf{v}) = \nabla \cdot (\mathbf{v}) = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z}$

Độ xoáy (Curl): $rot(\mathbf{v}) = \nabla \times \mathbf{v} = \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)\mathbf{a}_x + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)\mathbf{a}_y + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)\mathbf{a}_z$

Laplacian: $lapl(f) = \Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$

Véc-tơ chuyển dịch nguyên tố: $d\mathbf{l} = dx \cdot \mathbf{a}_x + dy \cdot \mathbf{a}_y + dz \cdot \mathbf{a}_z$, Thể tích khối nguyên tố: $dv = dx \cdot dy \cdot dz$

Hệ tọa độ cầu:

Gradient: $grad(f) = \nabla f = \frac{\partial f}{\partial r}\mathbf{a}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{a}_\theta + \frac{1}{r\sin\theta}\frac{\partial f}{\partial \varphi}\mathbf{a}_\varphi$

Độ tản: $div(\mathbf{v}) = \nabla \cdot \mathbf{v} = \frac{1}{r^2}\frac{\partial\left(r^2 v_r\right)}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial\left(\sin\theta v_\theta\right)}{\partial \theta} + \frac{1}{r\sin\theta}\frac{\partial v_\varphi}{\partial \varphi}$

Độ xoáy (Curl): $rot(\mathbf{v}) = \nabla \times \mathbf{v} = \frac{1}{r\sin\theta}\left(\frac{\partial\left(\sin\theta v_\varphi\right)}{\partial \theta} - \frac{\partial v_\theta}{\partial \varphi}\right)\mathbf{a}_r + \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial v_r}{\partial \varphi} - \frac{\partial\left(r v_\varphi\right)}{\partial r}\right)\mathbf{a}_\theta + \frac{1}{r}\left(\frac{\partial\left(r v_\theta\right)}{\partial r} - \frac{\partial v_r}{\partial \theta}\right)\mathbf{a}_\varphi$

Laplacian: $lapl(f) = \Delta f = \frac{1}{r}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \varphi^2}$

Véc-tơ chuyển dịch nguyên tố: $d\mathbf{l} = dr \cdot \mathbf{a}_r + r \cdot d\theta \cdot \mathbf{a}_\theta + r \cdot \sin\theta \cdot d\varphi \cdot \mathbf{a}_\varphi$,

Thể tích khối nguyên tố: $dv = r^2 \cdot \sin\theta \cdot dr \cdot d\theta \cdot d\varphi$

Hệ tọa độ trụ:

Gradient: $grad(f) = \nabla f = \frac{\partial f}{\partial s}\mathbf{a}_s + \frac{1}{s}\frac{\partial f}{\partial \varphi}\mathbf{a}_\varphi + \frac{\partial f}{\partial z}\mathbf{a}_z$ Độ tản: $div(\mathbf{v}) = \nabla \cdot \mathbf{v} = \frac{1}{\rho}\frac{\partial\left(\rho v_\rho\right)}{\partial \rho} + \frac{1}{\rho}\frac{\partial v_\varphi}{\partial \varphi} + \frac{\partial v_z}{\partial z}$

Độ xoáy (Curl): $rot(\mathbf{v}) = \nabla \times \mathbf{v} = \left(\frac{1}{\rho}\frac{\partial v_z}{\partial \varphi} - \frac{\partial v_\varphi}{\partial z}\right)\mathbf{a}_\rho + \left(\frac{\partial v_\rho}{\partial z} - \frac{\partial v_z}{\partial \rho}\right)\mathbf{a}_\varphi + \frac{1}{\rho}\left(\frac{\partial\left(\rho v_\varphi\right)}{\partial \rho} - \frac{\partial v_\rho}{\partial \varphi}\right)\mathbf{a}_z$

Laplacian: $lapl(f) = \Delta f = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho\frac{\partial f}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 f}{\partial \varphi^2} + \frac{\partial^2 f}{\partial z^2}$

Véc-tơ chuyển dịch nguyên tố: $d\mathbf{l} = d\rho \cdot \mathbf{a}_\rho + \rho \cdot d\varphi \cdot \mathbf{a}_\varphi + dz \cdot \mathbf{a}_z$, Thể tích khối nguyên tố: $dv = \rho \cdot d\rho \cdot d\varphi \cdot dz$